StoryCast: Research & Planning Document

Information Architecture, User Personas, Wireframes & WCAG 2.1 AA Conformance

Project Name: StoryCast Accessible Microsite	**Standard:** WCAG 2.1 Level AA
Tech Stack: Semantic HTML5, Sass, CSS Grid, Container Queries	**Date:** October 2026

1. Executive Summary & Purpose

StoryCast is an accessible, research-driven audio and video storytelling microsite engineered to eliminate digital barriers for people with sensory, motor, and cognitive disabilities. The site demonstrates how semantic HTML5, Sass partials with CSS cascade layers, and modern CSS container queries deliver state-of-the-art multimedia journalism without relying on client-side JavaScript frameworks.

2. User-Centered Research & Personas

Persona 1: Maya Chen (Deaf / Hard-of-Hearing Journalist)

Relies on verbatim closed captions and downloadable text transcripts. Frustrated by auto-generated captions with missed acoustic cues. StoryCast provides synchronized WebVTT captions, sound effect tags [echolocation click train], and full text transcripts.

Persona 2: David O'Connor (Screen Reader & Keyboard Specialist)

Low vision audio engineer navigating via NVDA/VoiceOver and keyboard only. Frustrated by missing skip links and unlabeled controls. StoryCast implements dedicated skip links, 3px high-visibility focus indicators, and semantic HTML5 buttons and landmarks.

Persona 3: Samira Patel (Commuter & Sensory Sensitive)

Neurodivergent graduate student with ADHD listening on noisy transit. Frustrated by unintelligible dialogue drowned by sound effects. StoryCast normalizes audio to -16 LUFS with high speech clarity and clickable transcript timecodes.

3. Information Architecture (IA) & Sitemap

The microsite consists of three semantically connected pages designed with clear visual and acoustic hierarchy:

• **Page 1: Home Page (/index.html):** Introduces StoryCast mission, features container-queried story cards, dynamic format filtering (Video, Audio, Field Recordings), and accessibility pillars.

• **Page 2: Story Detail (/story/echoes-of-the-deep.html):** Deep documentary immersion with HTML5 video, WebVTT closed captions, isolated hydrophone audio master (-16 LUFS), and an interactive synchronized transcript with real-time seeking and keyword search.

• **Page 3: About / Access (/about.html):** Conformance report, acoustic mastering specifications, keyboard navigation cheatsheet, and community accessibility feedback form.

4. WCAG 2.1 AA Accessibility Conformance Audit

Every component in StoryCast was audited against the four principles of WCAG 2.1 Level AA:

Principle	Criteria	StoryCast Implementation	Status
1. Perceivable	1.1.1 Non-text Content 1.2.2 Captions 1.4.3 Contrast	Descriptive alt text on all figures; WebVTT captions with sound cues; contrast ratios exceed 5.5:1 (up to 16.2:1 for body text).	**PASS (AA)**
2. Operable	2.1.1 Keyboard 2.4.1 Bypass Blocks 2.4.7 Focus Visible	All media & transcript controls keyboard operable; skip-to-content links; 3px high-contrast :focus-visible outlines on all targets.	**PASS (AA)**
3. Understandable	3.1.1 Language 3.2.3 Consistent Nav 3.3.2 Labels	<html lang="en"> declared; unified header/footer navigation with aria-current="page"; accessible form inputs.	**PASS (AA)**
4. Robust	4.1.2 Name, Role, Value 4.1.3 Status Messages	Valid semantic HTML5; ARIA live regions (aria-live="polite") for transcript filtering and status updates.	**PASS (AA)**

5. Technical Architecture & Container Queries

Sass Architecture: Modular partials organized under abstracts (_colors.scss, _typography.scss, _spacing.scss, _mixins.scss), base (_reset.scss, _globals.scss), layout (_grid.scss, _header.scss, _footer.scss), and components (_story-card.scss, _media-player.scss, _transcript.scss, _badges.scss, _buttons.scss).

CSS Cascade Layers: Explicitly ordered via @layer reset, tokens, base, layout, components, utilities to prevent specificity conflicts.

CSS Container Queries: Implemented on .c-story-card-wrapper via container-type: inline-size and container-name: storycard. At container widths >= 460px, the card morphs from a vertical stack into a side-by-side horizontal configuration. At >= 740px, it expands into a cinematic editorial banner.

6. Deliverable Links & Repository Structure

- **Interactive Wireframes & Mockups:** docs/mockups.html
- **Sitemap Diagram (SVG):** docs/sitemap.svg

- **Live Microsite:** index.html, story/echoes-of-the-deep.html, about.html
- **Compiled Styles:** css/main.css (compiled from sass/main.scss)
- **Media Assets:** assets/media/ (ocean-depths.mp4, echoes-audio.mp3, captions .vtt, transcripts .txt)